Bounded drafting in repeated games

Alisa Sheinkman

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Abstract

Does a bounded draft before an action improve an agent's decisions? In a completed 1,776-episode study using two NVIDIA Nemotron models on Nebius Token Factory, drafting lowered average exact benchmark loss, but worsened performance against Tit-for-Tat on average. Autonomous cooperation varied by pairing. Game Field combines inspectable play with a public action dataset and offline score reproduction. This amended study is descriptive, not a general model ranking.

1 Question and contribution

For students examining incentives and builders inspecting agents, cooperation counts alone do not establish decision quality. A model can earn more by cooperating less. We separate exact own-payoff loss against committed policies from outcomes between adaptive agents. The contribution is this inspectable Nemotron procedure comparison and its reproducible action dataset, not a first demonstration of LLM cooperation. Educational impact and real-world negotiation performance have not been evaluated.

2 Design

We compared direct action with a bounded draft-then-action procedure in two Nemotron models: Nano (30B-A3B) and Super (120B-A12B).[1] Players maximised expected own cumulative points in a repeated Prisoner's Dilemma: $CC = (3, 3)$, $CD = (0, 5)$, $DC = (5, 0)$, $DD = (1, 1)$. Choices were private until both players had acted.

Direct used one thinking-off action request (128-token cap). Draft used a thinking-on draft (1,024-token cap), then a separate thinking-off action request (128-token cap). Both received full completed history and used temperature 1 and top-$p = 1$. Only the current draft was added to the action request.

All 1,776 scheduled episodes completed in 24 balanced blocks:

- Track A: 1,536 episodes against disclosed AllC, AllD, Tit-for-Tat (TFT) or Grim policies. The 16 conditions covered continuation probabilities $\delta = 0.4, 0.6, 0.9$, one-round AllC/AllD, and ten-round TFT/Grim.
- Track B: 240 autonomous dyads covering all ten unordered pairings of the four model–procedure configurations, including self-play, at $\delta = 0.9$.

Four label/seat strata had six blocks each. Arms shared environment draws and stopping lengths within blocks; provider sampling randomness was not matched.

[1] Hosted IDs: `nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B` and `nvidia/nemotron-3-super-120b-a12b`.

3 Scoring

Track A loss measures expected payoff forgone against a committed opponent. The gap for action a_t at state s_t is $g_t = V^*(s_t) - Q^*(s_t, a_t)$, where continuation after that action is optimal. Episode loss is

$$L_{\text{geometric}} = (1-\delta)\sum_t g_t, \qquad L_{\text{fixed}} = \frac{1}{H}\sum_{t=1}^{H} g_t.$$

Geometric gaps are not discounted again: continuation is in Q^*, and termination supplies survival weighting. Lower loss is better; it measures neither action accuracy nor cooperation.

The primary contrast is draft minus direct mean loss within each model. The 16 conditions and four strata receive equal weight. Approximate 95% intervals use 20,000 stratified whole-block bootstrap draws.

Worked example. Against Always Cooperate in a one-round game, choosing C earns 3 points whereas D earns 5. The loss of C is therefore $5 - 3 = 2$; the loss of D is zero. This is foregone own payoff, not a moral judgement or cooperation score. A block repeats all conditions; a presentation stratum groups an action mapping and a player seat.

4 Results

Each procedure has 384 Track A episodes. Drafting reduced average loss for both models (Table 1), but the benefit varied by opponent (Table 2). The opponent breakdown is exploratory.

Model	Direct	Draft	Difference	Approx. 95% interval
Nano	0.6889	0.4942	−0.1947	[−0.3077, −0.0983]
Super	0.6150	0.3779	−0.2371	[−0.3095, −0.1735]

Table 1: Mean normalised oracle loss. Negative differences favour drafting.

Opponent	Nano difference	Super difference
AllC	−0.7562	−0.5146
AllD	−0.2135	−0.3708
Grim	−0.1575	−0.2542
TFT	+0.3485	+0.1913

Table 2: Draft minus direct loss, averaging four conditions per opponent.

Drafting worsened average performance against TFT. Against AllC and AllD, improvement means better selection of defection under the own-payoff objective, not more cooperation.

In Track B, mutual cooperation occurred in 189/2,060 rounds (9.2%); both defected in 1,042 rounds, and 829 were asymmetric. First-round mutual cooperation occurred in 26/240 dyads (10.8%). Rounds within a dyad are dependent; these counts do not supply 2,060 independent observations or a general ranking.

The pooled cooperation rate hides variation (Table 3). For example, Nano Direct self-play had 52/206 mutual-cooperation rounds, compared with 10/206 in Nano Draft self-play. These are descriptive comparisons under a shared stopping schedule; longer dyads contribute more to the round-level rate. The return normalisation is not the mean payoff per observed round.

Pairing	First CC / 24	CC / 206 rounds	Joint return
Nano Direct / Nano Direct	5	52	3.833
Nano Direct / Nano Draft	3	14	3.113
Nano Direct / Super Direct	5	33	3.529
Nano Direct / Super Draft	0	6	2.842
Nano Draft / Nano Draft	2	10	2.746
Nano Draft / Super Direct	3	19	3.271
Nano Draft / Super Draft	3	10	2.533
Super Direct / Super Direct	3	33	3.479
Super Direct / Super Draft	1	6	3.067
Super Draft / Super Draft	1	6	2.267

Table 3: All Track B pairings, 24 dyads each. Joint return is the mean of $(1 - 0.9)$ times the sum of both players' points over a dyad.

5 Interpretation and limitations

A best response to a commitment differs from an equilibrium [1]. Grim's initial cooperation threshold is $\delta = 1/2$; TFT's is $\delta = 2/3$. With ten announced rounds, either committed policy is best answered by nine cooperations then defection (32 own points). Finite-game backward induction for two strategic players does not imply immediate defection against these commitments. Geometric termination was hidden when choosing.

An earlier run stopped at 339/592 episodes after HTTP 502. Its outcomes were inspected before this successor, which retained the original 24-block schedule and excluded predecessor and engineering observations. These results are descriptive, not a new confirmatory preregistration.

Drafting changes compute, request count, thinking settings and conditioning together. Six blocks per stratum, one schedule, four policies and one provider window limit generalisation. Interval coverage and power are not established. There is no general model ranking or identified autonomous-play equilibrium; draft text does not verify internal beliefs.

6 Related work

Akata et al. [2] study LLM behaviour in repeated games and a social chain-of-thought intervention. Fontana et al. [3] examine cooperative behaviour against adversaries of varying hostility. Pal et al. [4] examine strategies, continuation probabilities, horizons and framing in repeated Prisoner's Dilemma. Our narrower extension compares bounded Direct/Draft procedures in two hosted Nemotron deployments using exact committed-opponent loss, alongside autonomous pairing outcomes. These results do not establish superiority over those studies' models, tasks or methods.

7 Execution and reproduction

The run used 17,124 successful requests, with no retries or protocol failures, in approximately 3 hours 8 minutes. Estimated final inference cost was $4.9278, not a provider invoice.

Drafting uses two requests per decision and longer inputs and outputs. Table 4 includes both benchmark and autonomous play, so it is not a Track A cost-effectiveness estimate. Public actions cannot independently validate usage aggregates or provider billing. The completed run used local CPU coordination and remote Token Factory inference; earlier cloud job probes failed before scientific inference.

The public app offers an action-by-action study replay and aggregate results. Its live educational mode instead uses temperature zero, reasoning requested off, and optional reported forecasts; it does not rerun this scientific protocol.

Procedure	Requests	Estimated USD
Nano Direct	2,854	0.068376
Nano Draft	5,708	0.795708
Super Direct	2,854	0.338590
Super Draft	5,708	3.725092

Table 4: Final evaluation costs across both tracks, priced at the recorded 20 October rates from retained usage aggregates.

Repository and public data: github.com/sheinkmana/game-field. The data package's executed plan and source index identify the study; current app and report revisions do not replace frozen execution evidence. Data and offline verification instructions are in `public/research/README.md`. Scores and intervals reproduce from revealed actions without inference. Raw responses, drafts and receipts remain private; public records cannot authenticate provider identity, raw transport or billing.

References

[1] Eric Rasmusen. *Games and Information: An Introduction to Game Theory.* Supplied fourth-edition draft, chapter headers dated 2005. Sections 1.2, 4.1, 5.1–5.2 and 6.5.

[2] Elif Akata et al. *Playing repeated games with Large Language Models.* arXiv:2305.16867, 2023; revised 2025.

[3] Nicolò Fontana, Francesco Pierri and Luca Maria Aiello. *Nicer Than Humans: How do Large Language Models Behave in the Prisoner's Dilemma?* arXiv:2406.13605, 2024.

[4] Saptarshi Pal et al. *Strategies of cooperation and defection in five large language models.* arXiv:2601.09849, 2026.